

Experiment No 2

Host Discovery

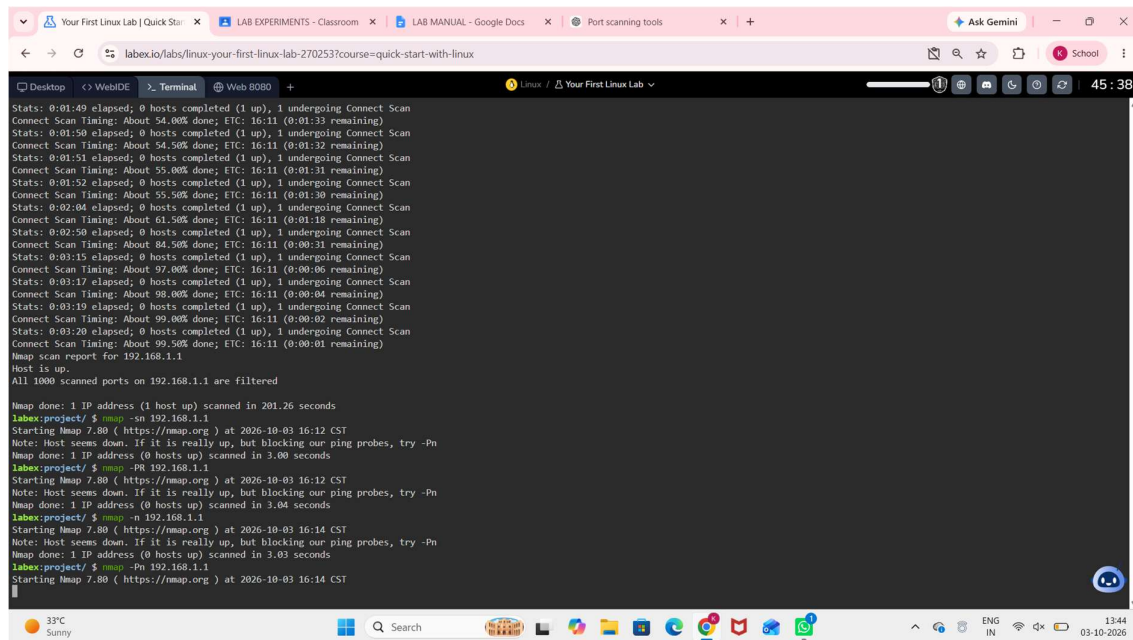
Aim

To study and perform **host discovery techniques using Nmap in Kali Linux** to identify active and reachable hosts in a network.

Algorithm

1. Start **Kali Linux** and open Nmap from **Applications** → **Information Gathering** → **Nmap**.
2. Identify an authorized target IP address or local network for testing.
3. Perform a **Ping Scan / Host Discovery** using:
`nmap -sn 192.168.1.1`
4. Perform a **No Ping scan** using:
`nmap -Pn 192.168.1.1`
5. Perform **ARP discovery** on the local network using:
`nmap -PR 192.168.1.1`
6. Disable DNS resolution and perform the scan using:
`nmap -n 192.168.1.1`
7. Observe the scan output and identify whether the target host is up or down.
8. Compare the results of the different host-discovery techniques.
9. Record the observations and terminate the scan.

Output



```
Stats: 0:01:49 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 54.00% done; ETC: 16:11 (0:01:33 remaining)
Stats: 0:01:50 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 54.50% done; ETC: 16:11 (0:01:32 remaining)
Stats: 0:01:51 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 55.00% done; ETC: 16:11 (0:01:31 remaining)
Stats: 0:01:52 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 55.50% done; ETC: 16:11 (0:01:30 remaining)
Stats: 0:02:04 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 61.50% done; ETC: 16:11 (0:01:18 remaining)
Stats: 0:02:30 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 84.50% done; ETC: 16:11 (0:00:31 remaining)
Stats: 0:03:15 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 97.00% done; ETC: 16:11 (0:00:06 remaining)
Stats: 0:03:17 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 98.00% done; ETC: 16:11 (0:00:04 remaining)
Stats: 0:03:19 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 99.00% done; ETC: 16:11 (0:00:02 remaining)
Stats: 0:03:20 elapsed; 0 hosts completed (1 up), 1 undergoing Connect Scan
Connect Scan Timing: About 99.50% done; ETC: 16:11 (0:00:01 remaining)
Nmap scan report for 192.168.1.1
Host is up.
All 1000 scanned ports on 192.168.1.1 are filtered

Nmap done: 1 IP address (1 host up) scanned in 201.26 seconds
labex:project/ $ nmap -sn 192.168.1.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-03 16:12 CST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.00 seconds
labex:project/ $ nmap -PR 192.168.1.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-03 16:12 CST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.04 seconds
labex:project/ $ nmap -n 192.168.1.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-03 16:14 CST
Note: Host seems down. If it is really up, but blocking our ping probes, try -Pn
Nmap done: 1 IP address (0 hosts up) scanned in 3.03 seconds
labex:project/ $ nmap -Pn 192.168.1.1
Starting Nmap 7.80 ( https://nmap.org ) at 2026-10-03 16:14 CST
```

Result

Thus, host discovery was successfully performed using Nmap in Kali Linux. The different options such as **-sn** (host discovery), **-Pn** (skip host discovery), **-PR** (ARP discovery), and **-n** (disable DNS resolution) were studied and their scan results were observed successfully.